

Terracotta E-Commerce Store: Project Explanation

1. Introduction

The Terracotta Store is a fully functional, dynamic e-commerce web application. It is designed with a premium, minimalist aesthetic using a curated "Terracotta" color palette and modern UI/UX principles. The application is built using a classic enterprise Java stack, ensuring robustness, scalability, and adherence to the MVC (Model-View-Controller) design pattern.

2. Technology Stack

- **Frontend:** HTML5, Vanilla CSS3 (Custom Properties/Variables, Flexbox, CSS Grid), JSP (JavaServer Pages), JSTL (JSP Standard Tag Library).
- **Backend:** Java Servlets (Jakarta EE).
- **Database:** MySQL.
- **Server:** Apache Tomcat.
- **Build Tool:** Apache Maven.

3. Architecture (MVC Pattern)

The application strictly follows the MVC design pattern to separate concerns:

- **Model:** Represents the data and business logic. It includes Java beans (POJOs) like `User.java`, `Product.java`, and `CartItem.java`. The Data Access Objects (DAOs) like `ProductDAO.java` and `UserDAO.java` handle database interactions.
- **View:** Responsible for presenting data to the user. This is handled by JSP files (`index.jsp`, `login.jsp`, `register.jsp`). JSTL is used to dynamically loop through data (like products) and render HTML.

- **Controller:** Java Servlets (`HomeServlet.java` , `LoginServlet.java`) act as controllers. They intercept HTTP requests from the View, communicate with the DAOs (Model) to fetch/update data, and forward the data back to the appropriate View.

4. Key Features & Implementations

Database & BLOB Image Streaming

Unlike traditional applications that store images as physical files on a server, this project implements an advanced database-driven media architecture. Product images are stored directly inside the MySQL database as binary data (`MEDIUMBLOB`). A dedicated `ImageServlet` is mapped to `/productImage` , which dynamically fetches the binary stream from the database and serves it directly to the browser as an image file. This ensures perfect data integrity and makes database backups complete with all media assets.

Dynamic Filtering Engine

The homepage features a dynamic filtering system. The application automatically queries the database for unique categories (`SELECT DISTINCT category...`) and renders the navigation bar. Users can filter by these categories or apply a maximum price constraint. The `ProductDAO` uses dynamic SQL query building to fetch only the matching records.

Advanced CSS & UI Engineering

The frontend uses highly advanced Vanilla CSS techniques:

- **Flex-Grow & Grid:** Ensures product cards stretch uniformly regardless of variable text length.
- **Mix-Blend-Mode:** Applied to product images to magically erase white backgrounds, seamlessly blending the products onto the Terracotta-colored cards.
- **CSS Animations:** Smooth `transform: translateY()` animations provide interactive feedback when hovering over products.

Data Migration & Image Processing Utilities

The project includes custom Java utility scripts (`ImageSlicer.java` and `CropDBImages.java`). These scripts leverage the `javax.imageio` library to mathematically slice large grid images

into individual product photos, crop borders, convert them to byte streams, and automate the database insertion process.

Short Summary

The Terracotta E-Commerce project is a secure, dynamic web application built on Java Servlets, JSP, and MySQL. It features user authentication, a cart system, and an advanced BLOB-based image streaming architecture. The frontend boasts a premium, responsive design powered by modern CSS, including dynamic database-driven categories, live price filtering, and elegant hover animations. The architecture adheres strictly to the MVC design pattern, ensuring clean, maintainable, and scalable enterprise code.